

Multiresolution Representation of the Background Medium

Haar MRA, Scale Refinement Inside a Krylov Solve,
and What May Legitimately Be Averaged

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Abstract

The proposal is to represent the background medium in a Haar multiresolution analysis, relaxing it to long scales during early GMRES iterates and reintroducing finer scales later. This note assesses it. Three findings. First, the *direction* conflicts with inexact-Krylov theory, which licenses accurate matrix–vector products *early* and relaxed ones *late* — the reverse; the idea belongs one level up, as nested iteration between solves rather than scale refinement within one Krylov space. Second, it has been done: [Hustedt et al. \(2004\)](#) built exactly this for frequency-domain seismic modelling in 2004, using a wavelet multiresolution to combine coarse- and fine-grid wavefields. Third, and most useful, a Haar average *is* the correct coarsening — but of the entries of the system matrix $\mathbf{A}$, not of (α, β, ρ) . Measured here: averaging $\mathbf{A}$ converges as $\mathcal{O}((H/\lambda)^2)$, while averaging velocities is 21× worse at the same wavelength and diverges as the wavelength shortens.

Contents

1	Haar multiresolution analysis	2
2	Where the refinement can go	2
2.1	Not inside one Krylov space	2
2.2	But nested iteration is exactly right	3
3	Prior art	3
4	What may legitimately be averaged	3
4.1	The question	3
4.2	The right variables are the entries of $\mathbf{A}$	3
4.3	Measured	4

5	The constraint from the coupling operator	4
6	Recommendation	5

1 Haar multiresolution analysis

A multiresolution analysis is a nested ladder of closed subspaces $V_j \subset V_{j+1}$ of L^2 with $\overline{\bigcup_j V_j} = L^2$, $\bigcap_j V_j = \{0\}$, the dyadic scaling property $f(x) \in V_j \Leftrightarrow f(2x) \in V_{j+1}$, and an orthonormal basis of V_0 generated by the integer translates of a scaling function φ (Mallat, 1989; Daubechies, 1992). The Haar case is $\varphi = \chi_{[0,1]}$, so V_j consists of functions piecewise constant on dyadic intervals of length 2^{-j} , with wavelet $\psi = \chi_{[0,1/2)} - \chi_{[1/2,1]}$ spanning $W_j = V_{j+1} \ominus V_j$ and two-scale relations

$$\varphi(x) = \varphi(2x) + \varphi(2x - 1), \quad \psi(x) = \varphi(2x) - \varphi(2x - 1). \quad (1)$$

Decomposition is running averages and differences: orthonormal, perfect reconstruction, $\mathcal{O}(N)$.

Two properties matter here. Haar has exactly *one* vanishing moment, so it represents piecewise-constant data exactly and smooth data poorly — an awkward fit for a background that is by construction the smooth depth average. And the refinement relation (1) is already familiar in this project: the subdivision identity used by the face-contact arbiter, splitting a cube into eight half-cubes with $\text{table}(n) = \frac{1}{8} \sum_m \text{mult}(m) \text{table}(m)$, is exactly (1) in three dimensions. The per-voxel contrast field is likewise a tensor-product Haar V_j at a single level.

2 Where the refinement can go

2.1 Not inside one Krylov space

The natural reading of the proposal — coarse operator early, fine operator late — runs against the established theory of inexact Krylov methods. Bouras & Frayssé (2005) and Simoncini & Szyld (2003) show that GMRES tolerates a perturbation $\Delta \mathbf{A}_k$ in the matrix–vector product at step k provided

$$\|\Delta \mathbf{A}_k\| \lesssim \frac{\eta}{\|r_{k-1}\|} \varepsilon, \quad (2)$$

so the admissible inaccuracy *grows* as the residual falls: be accurate early, sloppy late. The mechanism is that the perturbation enters the residual gap weighted by the Arnoldi solution coefficient $|y_k|$, and those decay as the iteration converges — an error committed in the first few basis vectors propagates into everything built on them. Reported gains are real but modest, 1.5–2.3× on Laplace and 2.7–3.3× on Stokes problems (van den Eshof & Sleijpen, 2004).

There is a second objection. If the operator itself changes between iterates, the space being built is no longer a Krylov space for a fixed operator and the minimal-residual property is lost, unless the variation is handled explicitly as a flexible (Saad, 1993) or inexact method.

2.2 But nested iteration is exactly right

The idea is sound one level up. Converge on the coarse medium, then use that solution as the *initial guess* for the finer one: nested iteration, or cascadic multigrid (Bornemann & Deuffhard, 1996), in which no coarse-grid correction is propagated back. This is between solves rather than within one, so neither objection applies. The same logic at the inversion level is multiscale full-waveform inversion (Bunks et al., 1995), where low frequencies are fitted before high — among the most influential ideas in the field.

3 Prior art

The specific construction has been built. Hustedt et al. (2004) present a multi-level direct-iterative solver for frequency-domain seismic modelling in which coarse- and fine-grid wavefields are combined either by space interpolation (DISS) or through the multiresolution scaling property of an orthogonal wavelet expansion (DIWS). They report that because the wavelet transform accounts for grid interactions, phase-shift artefacts are greatly reduced and significantly fewer iterations are required than with space interpolation.

One difference matters: they multiresolve the *wavefield and grid*, not the *medium*. Coarsening the medium is the harder problem, and it is where the real constraint lies.

4 What may legitimately be averaged

4.1 The question

A Haar projection of the background replaces fine structure by block averages. Of *what*? Averaging (α, β, ρ) over a block is the obvious choice and is wrong: for a finely layered medium the correct long-wavelength limit is Backus averaging (Backus, 1962), in which compliances average harmonically and the effective medium is transversely isotropic even when every constituent layer is isotropic. The general non-periodic statement is Capdeville’s homogenisation (Capdeville et al., 2010, 2018), where the scale separation is a spatial low-pass filter whose cutoff is set by the minimum wavelength and a precision factor, and the resulting smooth effective medium is what the waves actually see.

4.2 The right variables are the entries of $\mathbf{A}$

The layered problem marches as $\partial_z \mathbf{b} = \mathbf{A}(z) \mathbf{b}$, so a stack of thin layers has response $\prod_i \exp(\mathbf{A}_i h_i)$. As $h_i \rightarrow 0$ the Lie product formula gives

$$\prod_i \exp(\mathbf{A}_i h_i) \longrightarrow \exp\left(\sum_i \mathbf{A}_i h_i\right) = \exp(\langle \mathbf{A} \rangle H), \quad (3)$$

so the correct leading-order coarsening is the *arithmetic* average of the system matrix. This is not a coincidence: $\mathbf{A}$ is linear in

$$\gamma = \frac{\lambda}{\lambda + 2\mu}, \quad a = \frac{1}{\lambda + 2\mu}, \quad b = \frac{1}{\mu}, \quad \zeta = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad \chi = \frac{2\mu\lambda}{\lambda + 2\mu}, \quad \mu, \rho,$$

and these are precisely the combinations Backus averages arithmetically: $C_{33} = 1/\langle a \rangle$, $C_{44} = 1/\langle b \rangle$, $C_{66} = \langle \mu \rangle$, $C_{13} = \langle \gamma \rangle / \langle a \rangle$, $\rho_{\text{eff}} = \langle \rho \rangle$.

Haar averaging is therefore correct — in the variables of $\mathbf{A}$, and wrong in (α, β, ρ) .

4.3 Measured

A fixed realisation of a strongly contrasted fine layered medium (256 layers, $\alpha = 4.0 \pm 1.2$ km/s, $\rho = 2.6 \pm 0.4$ g/cm³, total thickness $H = 1$ km) is coarsened to a single homogeneous layer two ways and compared against the exact ordered product. The long-wavelength limit is approached by lowering the frequency, so the medium never changes and only H/λ varies.

f (Hz)	H/λ_p	velocity average	$\langle \mathbf{A} \rangle$ average
0.25	0.062	9.95×10^{-3}	4.72×10^{-4}
0.50	0.125	2.94×10^{-2}	1.78×10^{-3}
1.00	0.249	1.22×10^{-1}	9.34×10^{-3}
2.00	0.499	3.54×10^{-1}	3.01×10^{-2}
4.00	0.998	5.71×10^{-1}	1.58×10^{-1}
8.00	1.996	8.83×10^{-1}	1.29×10^{-1}

The $\langle \mathbf{A} \rangle$ average converges as $\mathcal{O}((H/\lambda)^2)$ — the error roughly quadruples as H/λ doubles — while the velocity average is 21× worse at the longest wavelength and degrades to complete failure within one wavelength.

What the residual is. Equation (3) is first-order Magnus; the leading correction comes from the non-commutativity of the $\mathbf{A}_i$, and it is exactly what Backus supplies in closed form as the non-linear term in $C_{11} = \langle \zeta \rangle + \langle \gamma \rangle^2 / \langle a \rangle$, which a plain arithmetic average of $\mathbf{A}$ cannot produce. In the realisation above that term is 13% of C_{11} , and it is what induces the anisotropy $C_{11}/C_{33} = 1.126$, $C_{66}/C_{44} = 1.148$. A multiresolution scheme that wants better than second order must carry it.

Caveat. The Backus constants quoted here are evaluated from the closed form; a transversely isotropic propagator was *not* independently validated in this exercise, so the anisotropy figures are analytic rather than measured.

5 The constraint from the coupling operator

There is a further, project-specific limit. The inter-scatterer coupling is extraordinarily sensitive to the reference: holding geometry fixed and varying only whether the crust reference carries

its depth gradient, a $\pm 0.5\%$ velocity gradient changes $G_9(i \leftarrow j)$ by 10%, and a realistic 67% gradient by 70–340% across frequency and slowness.

Coarsening the reference mid-solve is therefore *not* a small perturbation of $\mathcal{P}^z$. Whatever multiresolution scheme is adopted, that measurement sets the tolerance it must respect — and it is another reason to prefer nested iteration, where the operator is fixed within each solve, over scale refinement inside a single Krylov space.

6 Recommendation

1. Place the scale hierarchy *between* solves (nested iteration / cascadic), not within one Krylov space. If it must go inside, use FGMRES or an explicit inexact-Krylov bound, and note that (2) runs the opposite way to the intuition.
2. Coarsen the entries of $\mathbf{A}$, never (α, β, ρ) . Haar is then the right tool and is second-order accurate.
3. For better than second order, use Backus/Capdeville homogenisation, and accept that the effective medium is anisotropic.
4. For operator compression rather than medium representation, Haar’s single vanishing moment is inadequate for $1/r$ and $1/r^3$ kernels; use a basis with more vanishing moments in the Beylkin–Coifman–Rokhlin sense (Beylkin et al., 1991).
5. Note that the two-potential formulation is *already* a two-level scale split — smooth reference plus fine contrast. The natural generalisation is a hierarchy of such splits with coarse levels built by homogenisation.

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